

The S-matrix conjecture in every dimension $n \geq 158$

Automated mathematics run `fa_banach_001`
Lane 3, `agent_lane_03`, model GPT5.6

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Status. Candidate partial result, likely valid. A new projection estimate lowers the published cutoff from 1000 to 158. The even dimensions $4, 6, \dots, 156$ remain open.

Source conjecture

Let $\mathcal{D}_n$ be the set of real $n \times n$ matrices with entries in $[0, 1]$. An S-matrix is a $\{0, 1\}$ -matrix A satisfying

$$A^T A = \frac{n+1}{4}(I + \mathbf{1}\mathbf{1}^T).$$

Drnovšek [1] restates the Sloane–Harwit conjecture on PDF page 2 as follows: every nonsingular $A \in \mathcal{D}_n$ satisfies

$$\|A^{-1}\|_F \geq \frac{2n}{n+1},$$

with equality precisely for S-matrices.

2

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Conjecture. *If $A \in \mathcal{D}_n$ is a nonsingular matrix, then*

$$\|A^{-1}\|_F \geq \frac{2n}{n+1},$$

where the equality holds if and only if A is an S-matrix.

The source proves the conjecture for odd n , proves the sharper bound $\|A^{-1}\|_F \geq \sqrt{2}$ for $n = 2$, and obtains a weaker bound for even $n \geq 4$. Frankel and Urschel [2] later prove the conjecture for every $n \geq 1000$. We use their structural Lemma 3.2 verbatim and replace their fixed-parameter final step.

Result

Theorem 1. *If $A \in \mathcal{D}_n$ is nonsingular and $n \geq 158$, then*

$$\|A^{-1}\|_F \geq \frac{2n}{n+1}.$$

For even n the inequality is strict. Consequently the S-matrix conjecture holds in all odd dimensions, in dimension 2, and in every dimension $n \geq 158$.

Proof intuition

Frankel–Urschel show that a hypothetical even-dimensional counterexample is almost binary. From it one can select $t + 1$ binary rows and a short vector y . The first t row pairings with y must lie very close to $1/4$, while the last must lie close to 0. The selected rows have an exactly prescribed Gram matrix, so after orthonormalizing them, Bessel’s inequality says that the pairings cannot fluctuate that much unless y is longer than allowed.

Their proof fixes $t = 50$ and replaces the first two moments of the pairings by separate coarse bounds. The key improvement is to keep t free and write the completed-square error in terms of the *variance*. The t forced gaps between a quarter and zero then contribute directly. Optimizing the integer t makes the contradiction work from $n = 158$ onward.

The upstream structural lemma

Normalize a hypothetical counterexample B by $\|B\|_{\max} = 1$. For even n , integers $t \geq 4$ with $n \geq 4t$, Frankel–Urschel’s Lemma 3.2 gives $m = t + 1$, a matrix $C \in \{0, 1\}^{m \times n}$, and $y \in \mathbb{R}^n$ such that

$$CC^T = qI + p\mathbf{1}\mathbf{1}^T, \quad q = \left\lceil \frac{n}{4} \right\rceil, \quad p = \left\lfloor \frac{n}{4} \right\rfloor, \quad (1)$$

$$\|y\|_2^2 \leq \frac{1}{4(n-t)}. \quad (2)$$

Writing $z = Cy$, their coordinate estimates are

$$|z_j - \frac{1}{4}| < \varepsilon_+(n, t) \quad (1 \leq j \leq t), \quad (3)$$

$$|z_m| < \varepsilon_0(n, t), \quad (4)$$

where, with $s = n - t$,

$$\begin{aligned} \varepsilon_+(n, t) &= \frac{n-1}{4(n-2)n} + \frac{5}{4s} + \frac{1}{(8s-4)\sqrt{s}}, \\ \varepsilon_0(n, t) &= \frac{2n^2 - 4n + 1}{4(n-2)(n-1)n} + \frac{5}{4s} + \frac{1}{(8s-4)\sqrt{s}}. \end{aligned}$$

A free-parameter projection contradiction

Lemma 1. Set $m = t + 1$, $\gamma = n/q$,

$$d = \frac{1}{4} - \varepsilon_+ - \varepsilon_0, \quad \mu_- = \frac{t(\frac{1}{4} - \varepsilon_+) - \varepsilon_0}{m},$$

and

$$R = \frac{n}{4m(n-t)} - \gamma \frac{td^2}{m^2} - \gamma \left(1 - \frac{\gamma}{4}\right) \mu_-^2.$$

Assume $d > 0$, $\mu_- > 0$, $R \geq 0$, and

$$\frac{\gamma\mu_-}{2} - \sqrt{R} - \varepsilon_0 > \sqrt{\frac{n}{8(n-t)}}. \quad (5)$$

Then no C, y satisfying (1)–(4) exist.

Proof. The row-orthonormalization used in the proof of [2, Lemma 3.3] applies to any number m of selected rows. Put $\alpha = \mathbf{1}^T y$, and let

$$\mu = \frac{1}{m} \sum_{j=1}^m z_j, \quad \nu = \frac{1}{m} \sum_{j=1}^m z_j^2.$$

Bessel’s inequality and (2) give, by the same direct Gram-matrix calculation,

$$\alpha^2 - \gamma\mu\alpha + \gamma\nu \leq \frac{n}{4m(n-t)}.$$

Completing the square yields

$$\left(\alpha - \frac{\gamma\mu}{2}\right)^2 \leq \frac{n}{4m(n-t)} - \gamma \operatorname{Var}(z) - \gamma \left(1 - \frac{\gamma}{4}\right) \mu^2. \quad (6)$$

Here $\gamma \leq 4$. The coordinate estimates imply $\mu \geq \mu_- > 0$. Moreover,

$$\operatorname{Var}(z) = \frac{1}{m^2} \sum_{i < j} (z_i - z_j)^2 \geq \frac{td^2}{m^2},$$

because each of the first t coordinates differs from z_m by at least d . Thus the right side of (6) is at most R , and

$$\alpha \geq \frac{\gamma\mu_-}{2} - \sqrt{R}.$$

The last row of C contains $q + p = n/2$ ones by (1). Therefore

$$\alpha - z_m = \sum_{k: C_{m,k}=0} y_k, \quad |\alpha - z_m| \leq \sqrt{\frac{n}{2}} \|y\|_2 \leq \sqrt{\frac{n}{8(n-t)}}.$$

The lower bound for α , (4), and (5) give the strict reverse inequality, a contradiction. $\square$

Certified parameters and proof of Theorem 1

The interval checker `code/verify_cutoff.py` evaluates the hypotheses of Lemma 1 for every even n from 158 through 998 and searches the finite range $4 \leq t \leq \lfloor n/4 \rfloor$. It uses directed `mpmath.iv` arithmetic at 80 decimal digits and accepts a dimension only when the lower endpoint of the margin in (5) is positive. All 421 dimensions are certified. The weakest certificate occurs at $(n, t) = (160, 23)$ and has lower margin

$$0.0002579293062736265 \dots > 0.$$

The full transcript is in `verification.md`; the first choices include $t = 26$ for $n = 158$, $t = 23$ for $n = 160$, and $t = 26$ for $n = 162$.

Hence the structural consequence of a counterexample is impossible for every even $158 \leq n \leq 998$. Frankel–Urschel’s Theorem 1.2 handles all $n \geq 1000$, and the Drnovšek–Cheng theorem handles odd dimensions (with Drnovšek also treating $n = 2$). This proves Theorem 1. For even n equality is impossible because an S -matrix would have column weight $(n + 1)/2$, which is not an integer.

Verification, limitations, and novelty search

The code was rerun from the packet directory and checked all 421 dimensions; the final PDF and every rendered page were also inspected. The proof depends on the correctness of Frankel–Urschel Lemma 3.2, copied as a supporting PDF, but the new projection lemma is proved in full above.

The remaining unresolved scope is the finite list of even dimensions $4, 6, \dots, 156$. Seven focused upgrade routes were examined: free-parameter optimization, variance retention, directed-interval certification, sharpening the upstream constants, small-dimensional KKT/numerical analysis, dimension lifting, and inverse sign-pattern bounds. The last three do not supply a responsible finite-case proof, and the sharper tentative cutoff was not promoted because it modifies multiple upstream estimates.

A bounded search through 11 August 2026 used the exact conjecture name, the source and supporting titles/authors, citation trails, and threshold phrases. It found the published $n \geq 1000$ theorem but no prior cutoff below 1000. This supports, but does not prove, novelty. Human review should focus on the general- m Gram calculation in Lemma 1 and rerun the interval checker.

References

- [1] R. Drnovšek, *On the S -matrix conjecture*, Linear Algebra Appl. **439** (2013), 3555–3560; arXiv:1306.6786v2.
- [2] E. Frankel and J. Urschel, *On the Frobenius norm of the inverse of a non-negative matrix*, Linear Algebra Appl. **708** (2025), 193–203; arXiv:2409.04354.